

Citlalli Trejo Del Rio

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Full-stack developer with experience building scalable web applications, iOS apps, and machine learning systems. Proficient in Java, TypeScript, Swift, and Python with a focus on clean architecture, security, and end-to-end product development.

Education

University of Texas at Dallas 08/2024 - 06/2026

B.S. Computer Science

- **Coursework:** Machine Learning, Data Structures, Artificial Intelligence, Computer Vision, Advanced Algorithm Design and Analysis, Discrete Math, Calculus 2, Linear Algebra, Probability and Statistics, Computer Networks
- **Extracurricular:** Association for Computing Machinery; UTD Society of Automotive Engineers;

Experience

Top Escrow — Website Redesign Contractor 07/2025 – 08/2025

- Redesigned and rebuilt the company website using modern front-end practices, improving usability and visual consistency.
- Collaborated with stakeholders to deliver multiple revisions based on feedback.
- Supported launch by configuring domain setup and ensuring the site’s contact form successfully routed submissions.

Projects

Greige — Technical Co-Founder 05/2026 – Present

- Took a fashion resale marketplace from 0 → 1, owning the full Python (FastAPI) backend stack and various parts of the React (Next.js) frontend.
- Implemented end-to-end marketplace payments with Stripe Connect – programmatic seller onboarding/KYC, separate charge-and-transfer escrow, signature-verified webhooks, and transfer-reversal refunds.
- Architected a confidence-driven authentication pipeline combining a vision-language model (Claude), a DINOv2 visual-similarity model, and EXIF metadata analysis, with human-in-the-loop escalation for ambiguous cases.

[Pancake Money](#) — Technical Co-Founder 06/2025 – Present

- Led product vision and strategy for an iOS budgeting app, overseeing software development with a Python-based backend, AI tools, and Plaid API integration to simplify personal finance.
- Designed UI and interactive financial data visualizations in Swift, applying OOP principles to maintain a modular architecture. Conducted user research, competitive analysis, and iterative testing to inform feature prioritization.

Bigram Language Model & Autocomplete System

Tech Stack: Java 17, Maven, MySQL, JavaFX, JUnit 5

- Architected and implemented a bigram language model in Java using Kneser-Ney smoothing, supporting probabilistic text generation via beam search and greedy decoding.
- Designed a scalable MySQL persistence layer handling 600k+ relationships, optimizing ingestion with batched writes and idempotent upserts.
- Implemented a clean database abstraction layer (DBInterface) to support autocomplete and sentence generation features.

Phobos – Wishlist Web App

Tech Stack: Next.js, React, TypeScript, Node.js, Express, PostgreSQL, JWT, Bcrypt, Vercel, Railway

- Built a full-stack wish list platform with multi-page routing, collaborative sharing, and custom dark/light theme system.
- Developed a secure REST API with JWT authentication, bcrypt hashing, email verification, and account lockout protection.
- Designed with production readiness in mind, implementing rate limiting, input validation, and secure API practices.
- Deployed on Vercel and Railway under a custom domain, applying best practices including Helmet and CORS.

Partial Convolution U-Net Image Inpainting

Tech Stack: Python, PyTorch, TorchVision, OpenCV, Gradio

- Implemented a Partial Convolution U-Net (33M parameters) for irregular image inpainting based on published research.
- Achieved PSNR of 28–32 dB and SSIM of 0.92–0.96; evaluated model quality with FID using Inception V3 features.
- Built a full training pipeline with logging and an interactive demo interface.

Technical Skills

Languages: TypeScript, Python, Swift, HTML/CSS, JavaScript, Java

Frontend: React 18, Next.js 14 (App Router, Server Components, middleware), Tailwind CSS, Stripe Elements, responsive/accessible UI

Backend & Data: FastAPI (async), Celery, Postgres + pgvector, Redis, SQLAlchemy, Alembic, REST APIs, webhook design, data modeling

AI Systems: Multimodal/VLM integration, ensemble & human-in-the-loop pipelines, prompt and context engineering

Infrastructure: Git, GitHub Actions, Vercel, Railway, Docker